

Bracket Torque Assembly

Document No: WI-2201-A | Revision: C | Station: Final Assembly | Est. Cycle Time: 12 min

Required Tools

- Torque wrench, 5-25 Nm range
- 10mm socket
- 3mm hex key
- Anti-static wrist strap

Required Materials

- Mounting bracket, P/N BRK-4471
- M6x20 hex bolt (qty 4)
- M6 split lock washer (qty 4)
- Threadlocker, medium strength (blue)

Procedure

Step 1: Stage bracket on fixture

Place mounting bracket BRK-4471 onto assembly fixture F-12 with the chamfered edge facing up and toward the operator. Confirm the bracket seats flush against the fixture stops.

Step 2: Install bolts hand-tight

Insert one M6x20 hex bolt through each of the four mounting holes, each with a split lock washer underneath the bolt head. Thread each bolt in by hand until snug. Do not use a power tool for this step.

Step 3: Apply threadlocker

Remove bolt 3 (lower-right position) and apply one drop of medium-strength blue threadlocker to the threads before reinstalling hand-tight.

CAUTION: Threadlocker is a skin and eye irritant. Wear nitrile gloves and safety glasses. Avoid contact with painted surfaces.

Step 4: Torque bolts to spec

Using the torque wrench and 10mm socket, torque all four bolts in a star (criss-cross) pattern to the specification below. Record the final torque reading observed on the wrench.

Spec: Torque = 12 Nm $\pm$ 1 Nm (star pattern, all 4 bolts)

Step 5: Inspect bracket alignment

Visually confirm the bracket is flush and parallel to the reference edge of the fixture, with no visible gap greater than 0.5mm on any side. Mark this step Pass or Fail.

Step 6: Final visual inspection

Inspect all four bolt heads for damage, confirm no threadlocker residue is visible outside the joint, and confirm the bracket is free of scratches. Mark this step Pass or Fail.